

Inferring SMB from spaceborne observations.

Kejdi LLeshi,¹ G. Jouvét¹, M. Kneib², T. P.M. Leger¹, F. Herman¹

¹*Institute of Earth Surface Dynamics, University of Lausanne, Lausanne, Switzerland*

²*Department of Geography, University of Zurich, Zurich, Switzerland*

ABSTRACT. The distributed surface mass balance (SMB) of mountain glaciers is a key forcing for projecting glacier evolution and subsequent meltwater availability. Yet, direct in situ observations cover less than 0.2 % of the world’s glacierised area. Current approaches based on remote sensing struggle to disentangle the elevation change driven by SMB from that driven by dynamical thinning, particularly over the decadal observation windows during which the flux itself evolves. Here, we present a new SMB inference method, implemented within the open-source Instructed Glacier Model (IGM), that recasts the problem as a fully differentiable transient inversion. First, we perform an assimilation of observed surface ice speed data at a given time snapshot. Second, a forward simulation is then run between two epochs, whose digital elevation surfaces are known, under a candidate SMB. Automatic differentiation through the forward run yields the gradient of a geometry-misfit cost function with respect to the SMB, enabling efficient gradient-based optimisation of the latest. We apply the method to Argenti re Glacier in the French Alps over the period 2012–2021 and validate against the GLACIOCLIM stake network.. As a result, we obtain a root-mean-square error of $0.45 \text{ m w.e. yr}^{-1}$, outperforming the best results obtained with simpler approach. A sensitivity analysis shows Beyond the case study, our method is generic, computationally efficient and requires no glacier-specific tuning, making it readily transferable to any glacier with multi-temporal DEMs and surface-velocity data, and opening a path toward regional-scale SMB inversion.

INTRODUCTION

Mountain glaciers are sensitive recorders of climate and a critical component of regional water resources, yet projecting their future evolution requires knowledge of their distributed surface mass balance (SMB)—the spatial pattern of accumulation and ablation—which often remains poorly constrained at the global scale. Direct, in situ measurements rely on stake networks and snow pits, which are labour-intensive, hazardous in steep accumulation areas, and ultimately deployed on fewer than 0.2 % of the world’s glacierised area (Cogley and others, 2011). At the other end of the spectrum, regional and global climate models provide continuous gridded estimates of SMB, but their coarse resolution and process uncertainties limit their ability to reproduce glacier-scale mass changes without calibration. The vast majority of glaciers therefore remain effectively unmonitored, and regional-to-global mass-balance assessments depend on extrapolation from a small set of benchmark sites or on model-based reconstructions (Hugonnet and others, 2021; Zemp and others, 2019). As remote-sensing products of glacier surface velocity (Millan and others, 2022), ice thickness (Farinotti and others, 2019; Millan and others, 2022) and elevation change (Hugonnet and others, 2021; Beraud and others, 2022) reach near-global coverage, the question is no longer whether the data exist, but how to combine them to infer physically consistent SMB fields.

State-of-the-art

A first class of approaches derives SMB directly from climate models—regional climate models (RCMs) and Earth system models—that simulate surface energy balance and precipitation. These provide complete gridded SMB fields and have been widely used for large-scale assessments of glacier change, but their ability to reproduce observed glacier evolution is limited: modelled SMB must often be calibrated to match observed mass changes (Radić and others, 2014), comparisons with glacier-length records reveal systematic mismatches in the timing and magnitude of glacier fluctuations (Marzeion and others, 2012), and biases in precipitation phase, accumulation gradients and melt processes propagate into errors in simulated mass balance and geometry change (Rounce and others, 2020). These discrepancies stem from the coarse resolution of climate models and their simplified representation of the topographic and microclimatic effects that dominate high-mountain environments. SMB fields derived purely from climate models therefore rarely reproduce observed glacier retreat or advance without additional bias correction or data assimilation.

The standard remote-sensing route to distributed SMB instead exploits the mass-conservation equation,

which relates the rate of surface elevation change to the divergence of the ice flux and the SMB (Cuffey and Paterson, 2010). Given estimates of ice thickness and surface velocity, the vertically-integrated flux is computed and differentiated spatially to obtain the flux divergence; combined with an observed elevation-change field, this yields the SMB. Early implementations used flux gates to constrain emergence velocities in the ablation zone (Berthier and Vincent, 2012; Brun and others, 2018), and subsequent studies generalised the approach to fully distributed estimates over entire glaciers (Bisset and others, 2020; Miles and others, 2021; Pelto and Menounos, 2021; Van Tricht and others, 2021; Kneib and others, 2022). Numerical noise in the differentiation step is typically suppressed by spatial filtering or aggregation (Van Tricht and others, 2021; Bisset and others, 2020), at the cost of reduced accuracy and incomplete mass conservation.

The state-of-the-art application of this framework to a single glacier is that of Kneib and others (2024), who derived a high-resolution distributed SMB for Argenti re Glacier (French Alps) over 2012–2021 from Pl iades-derived elevation change and velocity, ground-penetrating-radar thickness measurements and three independent ice-thickness models. Validation against the GLACIOCLIM stake network yielded root-mean-square errors between 0.50 and 0.96 m w.e. yr^{−1} and correlation coefficients $R^2 > 0.91$, demonstrating that high-quality remote-sensing inputs can resolve spatial SMB patterns—including avalanche-fed accumulation hotspots—well beyond what stake networks alone can capture.

Yet a fundamental difficulty constrains all methods in this class: the observed surface elevation change reflects both the SMB and the flux divergence, and separating the two requires an independent estimate of one of them. In practice, the flux divergence is computed at a single observational snapshot and assumed representative of the entire observation period. Over decadal windows, however, glacier geometry, velocity and flux all evolve in response to ongoing climate forcing (J hannesson and others, 1989; Zekollari and Huybrechts, 2015), and treating the flux as stationary introduces a bias that grows with the length of the time-window. Kneib and others (2024) acknowledge this explicitly, noting that their estimates “could be refined to yearly or even seasonal timescales” if the stationarity assumption were relaxed. A second, practical limitation is the per-glacier nature of these workflows: smoothing length scales, density assumptions and weighting schemes are tuned site by site, which hampers transferability to the regional scales most relevant for water-resource applications.

Proposed solution

In this paper, we address these limitations by formulating SMB inference as an inverse problem solved entirely within a transient glacier-evolution model. We extend the open-source Instructed Glacier Model (IGM; ?) with a new module that, given an initial state calibrated by data assimilation (Jouvet, 2023), runs a transient forward simulation between two DEM epochs under a candidate SMB. Every operation of this forward run is tracked, so that automatic differentiation (Paszke and others, 2017) yields the full gradient of a geometry-misfit cost function with respect to the SMB. A gradient-based optimiser then iteratively updates the SMB until the simulated and observed glacier geometries agree. The SMB is parametrised as a function of surface elevation and read at every grid cell to produce a 2D SMB field; this regularises an otherwise ill-posed inverse problem (FH: I would probably give more explanations on why it solves the ill-posedness) and is appropriate for glaciers where elevation is the dominant control on SMB. By construction, the method does not assume a stationary flux: the flux divergence is recomputed at every time step from the evolving glacier state, so that the inferred SMB is the field dynamically consistent with the observed geometry change.

fh: i would the method first and then say you selected a site to illustrate and validate the approach (so introduce the study site then) In the following, we first describe the study site and the observational dataset (Sect. ??). We then detail the two-step pipeline—data assimilation of the initial state, followed by SMB inference by automatic differentiation (Sect.). We present the inferred SMB for Argentière Glacier over 2012–2021 and its validation against in situ stake measurements (Sect. ??), and finally discuss the implications for regional-scale SMB inversion (Sect. ??).

METHODS

Overview

We formulate the recovery of the surface mass balance (SMB) of a mountain glacier as a two-step pipeline implemented as a process module within the open-source Instructed Glacier Model (IGM; Jouvet 2023; Jouvet and Cordonnier 2023). In the first step, the parameters of the ice-dynamics model – the basal sliding coefficient field and the ice thickness field – are calibrated by data assimilation against an observational snapshot near the start of the inversion window. In the second step, this calibrated dynamical model is held fixed while a transient glacier-evolution simulation is run between two surface-DEM epochs and

differentiated end-to-end with respect to a parametrised SMB profile, allowing gradient-based inversion against the end-of-window thickness observation.

Ice-dynamics calibration via data assimilation

fh: i would give the equations already here and not wait the next setcion... The transient forward simulation introduced in Section couples the SMB field with the divergence of the ice flux, which itself depends on the ice thickness H and basal sliding coefficient c . A dynamical state in which H and c are mutually inconsistent produces unphysical mass redistribution that contaminates the SMB inversion, so the two fields must be calibrated jointly before the inversion is launched. We adopt IGM's existing data-assimilation module (Jouvet, 2023) for this step. fh: it would be useful to explain how the data assimilation actually works, even in layman terms

The module jointly inverts (H, c) by minimising the regularised misfit

$$\mathcal{J}_{\text{DA}}(H, c) = \alpha_u \|\mathbf{u}_s(H, c) - \mathbf{u}_s^{\text{obs}}\|^2 + \alpha_H \|H - H^{\text{obs}}\|_{\Omega_{\text{GPR}}}^2 + \mathcal{R}(H, c), \quad (1)$$

where $\mathbf{u}_s(H, c)$ is the modelled surface velocity returned by the IGM emulator (Jouvet and Cordonnier, 2023), the second term is evaluated only at the GPR survey points Ω_{GPR} (FH: the GPR data is appearing for the first time), and $\mathcal{R}$ is a Tikhonov-style smoothness regulariser on H and c . The minimisation is solved by L-BFGS; we refer the reader to Jouvet (2023) for the full formulation, weighting strategy and convergence behaviour. The output is a calibrated thickness H_{DA} and basal sliding field c_{DA} that together fix the ice-dynamics component of all subsequent forward simulations.

Construction of the observation target

The observation that drives the SMB inversion is the ice-thickness field at the end of the inversion window, T . We construct it from the calibrated bedrock and an end-of-window surface DEM. From the data-assimilation step the bedrock is fixed at

$$z_b(\mathbf{x}) = z_s^{\text{obs}}(\mathbf{x}, t_0) - H_{\text{DA}}(\mathbf{x}). \quad (2)$$

131 We assume z_b does not evolve over $[t_0, T]$ – which is justified on decadal timescales – and combine (2) with
 132 the observed end-of-window surface DEM $z_s^{\text{obs}}(\cdot, T)$ to obtain

$$H^{\text{obs}}(\mathbf{x}, T) = z_s^{\text{obs}}(\mathbf{x}, T) - z_b(\mathbf{x}). \quad (3)$$

133 Forward model: transient glacier evolution

134 With the dynamical state frozen, the glacier evolution between t_0 and T is governed by mass conservation,

$$\frac{\partial H}{\partial t} = b - \nabla \cdot \mathbf{q}, \quad \mathbf{q} = H \bar{\mathbf{u}}(H, z_s, c_{\text{DA}}), \quad (4)$$

135 where b is the surface mass balance, $\bar{\mathbf{u}}$ is the depth-averaged horizontal velocity returned by the calibrated
 136 emulator (Jouvet and Cordonnier, 2023), and $\mathbf{q}$ is the column-integrated ice flux. Because $\bar{\mathbf{u}}$ depends
 137 on the evolving thickness and surface elevation, the flux divergence is updated at every time step rather
 138 than held stationary as in flux-divergence-based geodetic methods (e.g., Miles and others, 2021; Kneib and
 139 others, 2024). Relaxing this stationarity assumption is the central methodological difference between the
 140 present approach and flux-divergence inversions, and is what allows the inversion to remain consistent over
 141 decadal observation windows during which the geometry, velocity and flux evolve appreciably.

142 Equation (4) is integrated explicitly with a forward-Euler scheme,

$$H^{n+1} = \max(0, H^n + \Delta t^n [b^n - \nabla \cdot \mathbf{q}^n]), \quad (5)$$

143 where the non-negativity projection prevents spurious negative thicknesses near the glacier margin. The
 144 time step is constrained by a Courant–Friedrichs–Lewy condition, $\Delta t^n = c_{\text{cfl}} \Delta x / \max_{\Omega} \|\bar{\mathbf{u}}^n\|$, with $c_{\text{cfl}} = 0.2$
 145 in our experiments and Δx the grid spacing. The flux divergence is computed by a centred finite-difference
 146 stencil on the staggered grid.

147 SMB parametrisation

148 The unknown to be inferred is a one-dimensional SMB profile $\beta(z)$, a function of elevation only, discretised
 149 at N uniformly spaced elevation bins by the values $\mathbf{m} = (m_1, \dots, m_N)$. At each time step of the forward
 150 simulation, this 1D profile is mapped to the two-dimensional source term entering (4) by interpolating β
 151 at the current surface elevation, $b(\mathbf{x}, t) = \beta(z_s(\mathbf{x}, t))$. The interpolation is piecewise linear and AD-friendly

– differentiable everywhere with respect to $\mathbf{m}$ – so gradients propagate cleanly through every time step. Reducing the inversion variable to a 1D elevation profile reflects the dominant first-order control on alpine SMB and reduces the dimensionality from $\mathcal{O}(10^4)$ pixels to $N = \mathcal{O}(10)$ bins, regularising an otherwise severely ill-posed inversion.

Inverse problem: SMB inference by automatic differentiation

We seek the profile $\mathbf{m}^*$ that minimises a misfit between the simulated end-of-window thickness $H^{\text{sim}}(\cdot, T; \mathbf{m})$ and the observation target $H^{\text{obs}}(\cdot, T)$ from (3). We define the cost

$$\mathcal{J}(\mathbf{m}) = \sqrt{\frac{1}{|\Omega|} \sum_{\mathbf{x} \in \Omega} (H^{\text{sim}} - H^{\text{obs}})^2} + \lambda \sum_{i=2}^{N-1} (m_{i+1} - 2m_i + m_{i-1})^2, \quad (6)$$

where the first term penalises geometry misfit and the second penalises the squared discrete second derivative of the profile, with λ controlling the trade-off between data fidelity and smoothness. The minimisation is performed with L-BFGS (?) using a memory of $M = 10$ correction pairs and a Hager–Zhang line search; the modest dimensionality of $\mathbf{m}$ makes the implicit Hessian approximation accurate and convergence is reached in $\mathcal{O}(10^2)$ function evaluations.

The gradient $\nabla_{\mathbf{m}} \mathcal{J}$ is obtained by reverse-mode automatic differentiation through the entire forward integration (4)–(5). Every operation in the forward map – the emulator evaluation, the staggered-grid divergence, the explicit Euler update, the elevation-based interpolation, and the cost itself – is implemented as a differentiable TensorFlow tensor operation, so that the chain rule can be applied uniformly across the entire forward graph.

The principal computational limitation is memory. A naïve reverse-mode AD trace stores the full computational graph of the forward integration, whose memory footprint scales as $\mathcal{O}(N_t \cdot N_\Omega)$, with N_t the number of time steps and N_Ω the number of grid points; for the Argenti re case study this exceeds the memory budget of a single GPU. We address this with gradient checkpointing (?): the forward integration is partitioned into K segments, only the segment-boundary states are stored during the forward pass, and the within-segment graphs are reconstructed by re-running the forward integration during the backward pass. With $K = \sqrt{N_t}$ segments the memory cost reduces from $\mathcal{O}(N_t)$ to $\mathcal{O}(\sqrt{N_t})$ at the price of one additional forward evaluation per backward pass.

Implementation

The full pipeline is implemented as the `smb_inference` process module of IGM. The module reads a calibrated state produced by IGM’s `data_assimilation` module, builds the forward integrator with the calibrated dynamics held fixed, and runs the L-BFGS inversion defined in (6). All hyperparameters are exposed in the Hydra-based configuration system and listed in Table ??.

RESULTS

Overview

We validate the proposed framework on Argenti re Glacier (French Alps), one of the most densely monitored mountain glaciers in the world. The glacier is covered by the long-running GLACIOCLIM stake network, has been surveyed by repeated ground-penetrating-radar (GPR) campaigns (Rabatel and others, 2018), and is observed by an exceptionally rich Pl iades archive from which both surface elevation change and surface velocity are available at high spatial resolution (Beraud and others, 2022; ?). This dataset has recently been exploited by Kneib and others (2024) to produce a state-of-the-art distributed SMB reconstruction over the period 2012–2021, against which we benchmark our results below.

We apply our pipeline to the same observational dataset and over the same period, with one deliberate restriction: we exclude the Tour Noir glacier from the analysis. Tour Noir is no longer dynamically connected to the main trunk of Argenti re, so it neither receives nor delivers ice flux from the rest of the system, and including it as a disconnected patch would degrade the inversion without informing it. The Am thystes tributary, which remains attached to the main trunk, is retained throughout. We also note that the inversion does not represent avalanche redistribution, a process for which Kneib and others (2024) inferred local precipitation correction factors of $P_{\text{fact}} \approx 1.6$ at the base of steep adjacent headwalls; the implications of this choice are taken up in Sect. ??.

The findings that follow are organised in three parts. We first describe the calibrated ice-dynamics state recovered by the data-assimilation step (Sect.). We then present the SMB profile inferred by automatic differentiation through the transient forward run, together with diagnostics of the forward-model consistency (Sect.). Finally, we validate the inferred SMB against the in situ GLACIOCLIM stake measurements and benchmark its performance against the three modelling approaches of Kneib and others (2024) (Sect.).

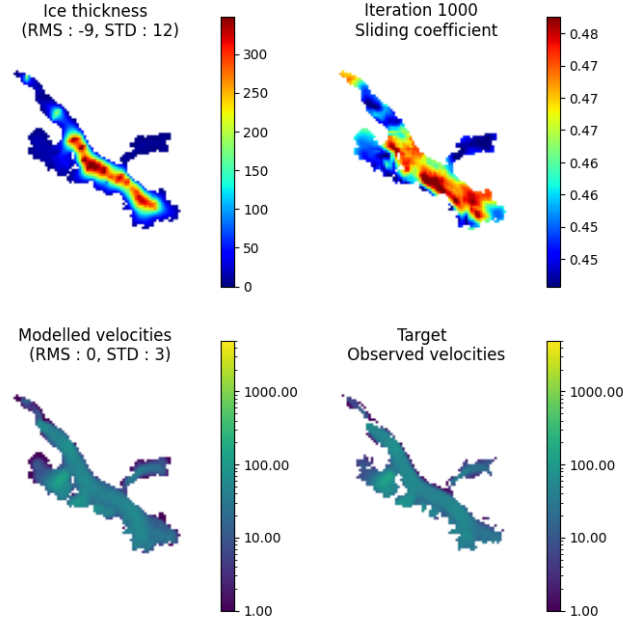


Fig. 1. Output of the data-assimilation step on Argentière Glacier. Top row: calibrated ice thickness H_{DA} , calibrated basal sliding coefficient c_{DA} , and residual top-surface adjustment. Bottom row: modelled surface velocities, observed surface velocities, and the resulting flux divergence $\nabla \cdot \mathbf{q}$ at $t_0 = 2012$. Modelled velocities reproduce the observations with $RMS = 3 \text{ m yr}^{-1}$; the calibrated thickness departs from the GPR transects by $RMS = 12 \text{ m}$.

Calibrated ice-dynamics state

The data-assimilation step jointly retrieves the ice-thickness field H_{DA} and the basal-sliding coefficient field c_{DA} that, together, reproduce the observed surface velocity and the GPR thickness transects. The inversion converges within 10^3 L-BFGS iterations and yields a self-consistent dynamical state.

Modelled surface velocities reproduce the observations with a root-mean-square residual of 3 m yr^{-1} , an order of magnitude below typical trunk velocities ($20\text{--}80 \text{ m yr}^{-1}$, locally exceeding 150 m yr^{-1} on the Rognons confluence). The calibrated thickness departs from the GPR transects by $RMS = 12 \text{ m}$ with a mean bias of -9 m , of the order of the GPR-derived thickness uncertainty itself (Rabatel and others, 2018). The calibrated sliding coefficient shows the expected spatial structure, with enhanced sliding along the central flowline and reduced sliding near the lateral margins (Fig. 1).

This calibrated (H_{DA}, c_{DA}) pair fixes the bedrock elevation through Eqn. (??) and defines the frozen dynamical substrate on which the transient forward simulation of Sect. is run. Integrating this state forward for one year under a zero-SMB forcing produces no spurious thickness drift, confirming that the dynamical state is in mechanical equilibrium with itself and that any subsequent geometric evolution must therefore be attributable to the prescribed SMB rather than to residual model imbalance.

Inferred SMB and forward-model consistency

Holding $(H_{\text{DA}}, c_{\text{DA}})$ fixed, we run the transient forward simulation from 2012 to 2021 and invert for the elevation-resolved SMB profile $\beta(z)$ that minimises the geometry-misfit cost (Eqn. ??). The L-BFGS optimiser converges in approximately 10^2 function evaluations, with the cost dropping by more than two orders of magnitude before stagnating at a level consistent with the combined noise floor of the DEM differencing and the GPR-derived bedrock.

The retrieved profile $\beta(z)$ is shown in Fig. 2. It is monotonically increasing across the full altitudinal range of the glacier (1670–3610 m a.s.l.), with strongly negative values at the tongue (reaching $-8.5 \text{ m w.e. yr}^{-1}$ at 1700 m) and a positive accumulation regime in the upper basin. The simulated end-of-window thickness $H^{\text{sim}}(\cdot, T)$ matches the observation target $H^{\text{obs}}(\cdot, T)$ everywhere on the trunk to within the DEM differencing uncertainty, with no systematic spatial bias.

A central diagnostic of the method is that the flux divergence is recomputed at every time step rather than held stationary, as is the standard assumption in geodetic SMB inversions (Kneib and others, 2024; Miles and others, 2021). Figure 3 compares the modelled $\nabla \cdot \mathbf{q}$ field at the start (2012) and end (2021) of the inversion window. The two snapshots differ substantially: the 2021 divergence pattern is everywhere weaker in magnitude than its 2012 counterpart, with the largest changes localised in the ablation zone where the glacier has thinned most. Treating either snapshot as representative of the entire 2012–2021 window — as is implicit in single-snapshot flux-divergence inversions — would therefore introduce a bias whose magnitude scales with the geometric change accumulated over the observation period.

Quantitatively, the root-mean-square magnitude of the flux-divergence field decreases from $\text{RMS}_{2012} = ?? \text{ m yr}^{-1}$ to $\text{RMS}_{2021} = ?? \text{ m yr}^{-1}$, a reduction of ?? % over the inversion window. The RMS of the pointwise difference between the two fields, $\sqrt{\langle (\nabla \cdot \mathbf{q}|_{2021} - \nabla \cdot \mathbf{q}|_{2012})^2 \rangle} = ?? \text{ m yr}^{-1}$, is itself comparable in magnitude to the SMB signal that single-snapshot inversions seek to recover, indicating that the bias incurred by treating the flux as stationary is of the same order as the quantity being inferred. Argentière is, moreover, a relatively favourable case: it is geometrically modest in extent, drains a single dynamically-coherent catchment, and lost only a few tens of metres of thickness over the decade. On larger glaciers, on longer observation windows, or in regions undergoing faster geometric adjustment, the divergence between the start- and end-of-window flux fields can only grow, and with it the case for the transient treatment adopted here.

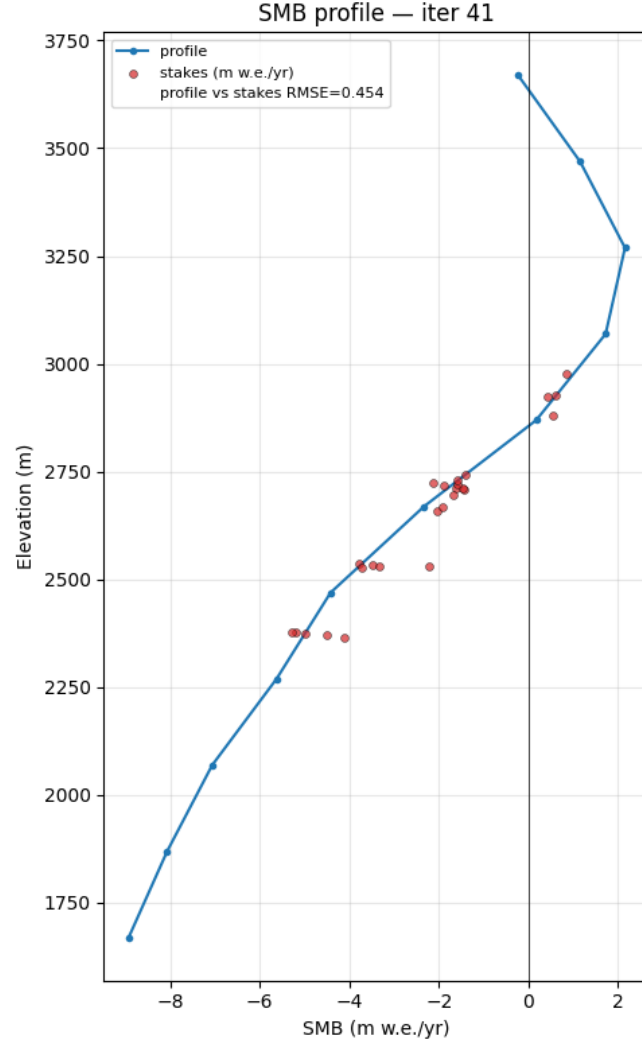


Fig. 2. Inferred elevation-resolved SMB profile $\beta(z)$ (solid line) for Argentière Glacier over 2012–2021. Red dots show the mean annual GLACIOCLIM stake measurements over the same period, plotted at the elevation of each stake. The profile is sampled at 11 elevation bins of 200 m width spanning 1670–3610 m a.s.l.

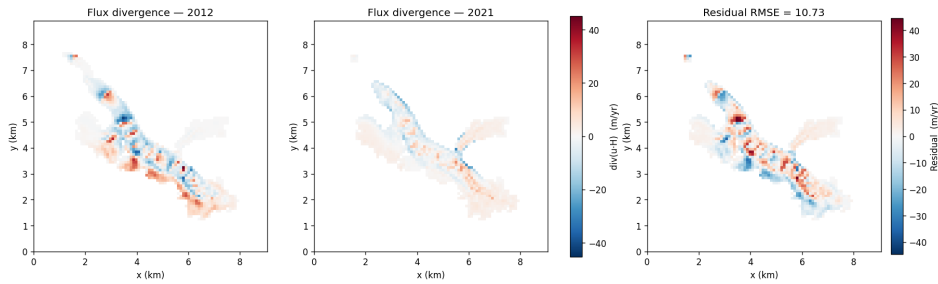


Fig. 3. Modelled flux divergence $\nabla \cdot \mathbf{q}$ on Argentière Glacier at the start (left, 2012) and end (right, 2021) of the inversion window. The two fields differ considerably in both magnitude and spatial pattern, in contradiction to the stationary-flux assumption underpinning single-snapshot flux-divergence SMB inversions.

Table 1. Validation of inferred SMB against the GLACIOCLIM stake network on Argenti re Glacier (main trunk + Am thystes tributary) over 2012–2021. The three baselines are the modelling approaches of Kneib and others (2024), evaluated against the same set of stakes.

Method	RMSE [m w.e. yr ^{−1}]
F2019 (Kneib and others, 2024)	0.50
SIA (Kneib and others, 2024)	0.67
IGM-distributed (Kneib and others, 2024)	0.85
Ours (1D AD inversion)	0.454

Validation against the GLACIOCLIM stake network

We sample the inferred SMB profile $\beta(z)$ at the elevation of each GLACIOCLIM stake located on the main trunk of Argenti re Glacier and the Am thystes tributary, and compare the resulting values to the in situ stake measurements averaged over 2012–2021. The inferred profile reproduces the observed altitudinal pattern across the full elevation range of the stake network (Fig. 2).

The aggregate fit, summarised in Table 1, yields a root-mean-square error of $\text{RMSE} = 0.454 \text{ m w.e. yr}^{-1}$ between the inferred and measured SMB. This value is lower than that of every previously published distributed SMB reconstruction on this site over this period: the F2019, SIA and IGM-distributed approaches of Kneib and others (2024) report RMSEs of 0.50, 0.67 and 0.85 m w.e. yr^{−1}, respectively, against the same stakes.

We emphasise that this is not a pixel-by-pixel comparison. Our inversion variable is one-dimensional by design: at any given elevation, the inferred SMB is the value that, averaged over the glacier surface at that altitude, is most consistent with the end-of-window thickness observation. Local horizontal heterogeneity at a given elevation — in particular, the avalanche-fed accumulation hotspots at the base of the headwalls reported by Kneib and others (2024) — is therefore not resolved by the inversion. This is a deliberate compromise. Distributed SMB inversion is severely ill-posed: a fully 2D SMB field has dimensionality comparable to the number of grid cells, while the geometric constraint provides at most one scalar per cell, and that constraint is uninformative wherever the ice is absent or thin. Reducing the inversion to a 1D elevation profile collapses this dimensionality from $\mathcal{O}(10^4)$ to $\mathcal{O}(10)$, regularising the problem to a degree that does not require dense GPR coverage or per-site smoothing length scales — prerequisites that limit the regional transferability of fully distributed approaches. The trade-off this entails, and the conditions under which the 1D restriction is worth paying, are taken up in Sect. ??.

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